



Investigation report

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads - Standard investigation

Drafted 04/10/2026

INVESTIGATION WORKFLOW SUITE

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Executive summary

What happened

On 07/07/2026, MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers, grounded on Boolarong Bank at 18:12:04 while inbound in darkness on service TF18 through the Weerona Channel at Kanooka Heads. Thirty-eight passengers, a master, a deckhand and a kiosk attendant were on board. The vessel struck the bank at 6.8 kt and stopped 74 m west of the charted centreline and 54 m outside the dredged channel, on a falling tide.

No one was seriously injured. One passenger reported a grazed shin and two were treated at the wharf for distress. The starboard hull plating split over the No. 2 void, which flooded progressively, and the starboard propeller, shaft, rudder stock and skeg were damaged. All passengers were ashore by 19:26. The vessel was refloated at 00:20 and withdrawn from service, and the channel was closed to commercial traffic for 19 hours.

What the investigation found

At 18:07:06 the chart plotter, the passage plan's primary means of position fixing, froze and did not recover. The master continued at about 8 kt searching for the No. 4 beacon, the secondary means, which mangrove canopy on Weerona Islet screened from the channel throughout. The heading was increased by about five degrees and held, and the vessel crossed the channel's western edge 114 seconds after the freeze, although stopping was possible throughout on flat water with no other traffic. The echo sounder's shallow alarm, set at 0.9 m under the keel since 2019, gave 16 seconds' warning against a crash stop distance of 92 m from 8 kt.

The operator had no night passage plan for the channel and no hold point for loss of the primary means of fixing. This very likely contributed. The winter timetable change that moved TF18 into darkness was assessed as a timetable change only, and the route risk assessment restricting the channel to daylight was not reopened. This likely contributed.

The plotter had frozen twice that morning, but the fault was passed on verbally at the change of master, as the operator's handover practice permitted, rather than entered in the defect book. Firmware for the fault issued in 11/2024 had never been applied. This likely contributed. A written entry would not alone have stopped TF18, so this is also a finding about fleet defect reporting.

Clearing of the beacon's sightline had been deferred unfunded for three works rounds because the islet lay outside a 50 m scope. This very likely contributed. The beacon's solar conversion halved its nominal range without any operator being told, and the passage plan still gave 6 M. This likely contributed. The waterways code's availability and notification duties do not apply to this channel. This likely contributed, although this finding is less settled: the lantern was lit throughout, and no availability figure would have detected screening.

Whether new foreshore floodlighting behind the beacon affected the master's ability to see it could not be determined. The floodlights and beacon were aligned, but the only measured alignment lay inside the screened arc.

Two matters increased risk without influencing this occurrence. The forward saloon lifejacket locker could not be reached from 22 of 34 seats, and the winter roster places darkness transits at the end of long split duties. After contact, the master's decision not to refloat the holed vessel reduced risk.

Safety issues and actions

Safety issues were identified in the operator's change management, night passage planning and defect handover, and in how aids to navigation on this channel are maintained, scoped and notified. The lifejacket locker arrangement, created by a 2019 refit that did not trigger a re-survey, is also a safety issue, though not contributing.

The canopy was trimmed to 2 m on 24/07/2026 under a funded three-yearly cycle. Local notice LNM-2026-14 gave the beacon's 3 M range, and the operator reissued the passage plan appendix as issue 3. The investigation has recommended that the services agreement replace the 50 m vegetation scope with a sightline test, and that Tarrakin Ferries write a night passage plan, naming loss of the primary means of fixing as a hold point, before any further darkness transit.

Event overview

On the evening of 07/07/2026, MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers, was operating service TF18, the last service of the day from Muntari Island to the town wharf at Kanooka Heads. TF18 was a winter timetable evening return, made in darkness every day since 01/06/2026, and this was the thirty-first such service. The crew were a master, a deckhand and a kiosk attendant.

The master took over the vessel at about 13:05 at the town wharf. Both masters' accounts of the handover agree. The outgoing master said the chart plotter had frozen twice that morning and recovered by itself both times. The relieving master replied that it had done the same to him the month before. The plotter's event log recorded watchdog resets, each with cause code 0x51, at 10:42 and 11:17 local. The outgoing master recalled that neither freeze happened inside the Weerona Channel. Neither master entered the fault in the wheelhouse defect book, which held no entry for 07/07/2026 before the occurrence. The two afternoon round trips were otherwise unremarkable.

The voyage

Sunset was at 17:02 and civil twilight ended at 17:29. The master described clear conditions, a light westerly and flat water. His pre-departure public address noted that it was dark and "a bit fresh out on the aft deck". TF18 departed Muntari Island at 17:35:02 with 38 passengers and reported to Port Control on VHF channel 12, which replied "nothing to affect you, channel's clear".

At 18:03:58 the vessel entered the Weerona Channel from the north, abeam the No. 1 beacon, at 8.4 kt on a course over the ground of 195 degrees. At about 18:04 the deckhand asked whether the master wanted her to stay on the bridge. He replied, "Nah, go and do the fares". She recalled being in the saloon from about 18:05 for the rest of the transit, leaving the master alone on the bridge.

The channel transit

At 18:05:30 the vessel was abeam the No. 2 beacon, 1 m west of the charted centreline and 0.90 nm north of the No. 4 beacon. The master was heard to say "Two ... good." Both engines were running steadily at 1,780 and 1,784 rpm, with no alarms and no throttle movement.

At 18:07:06, 0.68 nm north of the No. 4 beacon, the plotter's event log recorded that the display process had stopped responding, with the same cause code as that morning, but no reset followed. The master said the chart stayed on the screen with the vessel symbol stationary and the display would not accept touch input. The bridge audio captured "Oh, come on" at 18:07:11, sounds consistent with repeated screen taps at 18:07:26, and "frozen again" at 18:07:44. The display did not recover, and power was removed from the unit at 18:16.

The master continued at the same speed and looked for the No. 4 starboard-hand beacon on the western side of the channel, the established visual reference on that route. From 18:07:06 the heading increased steadily from 195 to 200 degrees over about a minute and was then held. Rate of turn peaked at 2.1 degrees per minute at 18:07:40 and returned to zero by 18:08:50. Steering gear hydraulic pressure stayed within its normal band, and rudder angle followed helm demand within 1.5 degrees. The master stated that he had eased the vessel about five degrees to starboard while searching for the beacon, without deciding to do so.

At 18:08:31, 12 m west of the centreline and 0.47 nm north of the beacon, the master said "Right. Where's four." At about 18:09, 0.40 nm north of the beacon, the vessel crossed the western edge of the dredged channel at 8.0 kt, then moved west at an average of 12 m per minute. At 18:09:40

the audio recorded a sound consistent with a window or door sliding open. The master said he opened the wheelhouse window in case the glass was the problem.

At 18:10:02, 39 m west of the centreline and 0.25 nm north of the beacon, the master said "There's four. Keep her over." He recalled that the beacon appeared dead ahead rather than on the starboard bow, and that he understood at once he was on the bank side of the channel. A range and visibility check on 18/07/2026 found the beacon could not be seen from the channel between 0.84 nm and 0.34 nm north of it, because the sightline passes through the mangrove canopy on Weerona Islet. Both masters also described the beacon as hard to pick out at night against the glow of the town and the new car park floodlights, though the master could not say whether the lights were why he could not see it.

By 18:11:00 the vessel was 58 m west of the centreline and over Boolarong Bank. Speed over the ground had fallen from 8.4 kt to 7.6 kt since 18:07 with no throttle movement, consistent with shoaling water beneath the hulls.

The grounding

The echo sounder's shallow alarm, set at 0.9 m under the keel since commissioning in 2019, sounded at 18:11:48, with the vessel 70 m west of the centreline at 7.1 kt. The master said "No no" at 18:11:50, and at 18:11:52 both throttles were moved to astern together. Before astern thrust reached full, at 18:12:04, the vessel made contact with Boolarong Bank at 6.8 kt, having covered 58 m in the 16 seconds after the alarm. The vessel's trials figure for a crash stop from 8 kt is 92 m.

The audio recorded a heavy impact, then grinding and vibration for approximately 9 seconds. The vessel stopped within 11 m, heeled 4 degrees to starboard, at a surveyed position 74 m west of the charted centreline, 54 m outside the dredged channel and 34 m west of the No. 4 beacon. The bed there was 0.50 m below chart datum, and the tide was falling. The master stopped both engines at 18:12:31 and did not restart them.

The response

At 18:12:40 the master told passengers that the vessel had touched the bottom and was stopped. At 18:13:04 he broadcast a pan pan to Kanooka Heads Port Control on VHF channel 16, giving position, persons on board and no reported injuries or pollution, stating he was not attempting to refloat, and requesting the workboat for a passenger transfer. Port Control acknowledged and called out the workboat. The master cited the falling tide, the unknown bottom and the number of people aboard for not refloating.

At 18:14:52 he asked passengers to put on lifejackets. The deckhand worked aft to forward. In the forward saloon the locker was behind the kiosk servery counter, and she had to ask three people standing there to move and pass through a narrow gap at its end. She reported everyone in a lifejacket at about 18:22.

At about 18:16 the deckhand sounded the No. 2 starboard void and found water. At 18:17:05 the master reported slow ingress of approximately 75 mm to Port Control, monitored every ten minutes. The workboat came alongside the port quarter at 18:41, and at 19:26:08 the master reported all 38 passengers ashore. One passenger reported a grazed shin and two were treated at the wharf for distress. No one was seriously injured.

The vessel was refloated on the rising tide at 00:20 on 08/07/2026, with approximately 260 mm of water in the No. 2 void, and towed to the town wharf. The starboard hull bottom plating was set up over 4.1 m, with a 180 mm split in way of the No. 2 void. The starboard propeller blades and shaft were bent, the starboard rudder stock was distorted 3 degrees, and the starboard skeg was

torn away. The port hull, propulsion and steering were undamaged. The vessel was withdrawn from service, and the Weerona Channel was closed to commercial traffic for 19 hours.

Analysis

Introduction

This analysis follows the sequence from the decisions and conditions furthest from the bridge to the moment of contact. Each thread is placed under the heading where it belongs. Contributing safety factors are distinguished from other factors that increased risk without influencing this occurrence. One matter that was examined could not be settled, and it is discussed where it arises.

People

Departure from the dredged channel

The vessel did not drift out of the Weerona Channel. The AIS track and the vessel's recorders show it on the charted centreline abeam the No. 2 beacon at 18:05:30, 1 m west of it, and holding there until 18:07:06. From that time the heading increased steadily from 195 to 200 degrees over about a minute and was then held. The rate of turn peaked at 2.1 degrees per minute at 18:07:40 and was back to zero by 18:08:50.

That shape is a heading change applied and then held. It is not the signature of a current or wind effect. The wind was offshore at 8 kt, and there is no cross-set in the channel at that state of the tide. On the new heading the vessel set west at an average of 12 m per minute. It reached the channel's western edge at about 18:09, was over Boolarong Bank from about 18:11, and came to rest at a surveyed position 74 m west of the centreline.

The channel is 40 m wide and dredged to 3.2 m. The vessel grounded only because it was outside the channel. Every factor upstream of this point matters because it explains why the vessel came to be there.

It is virtually certain that the vessel left the dredged channel and tracked onto Boolarong Bank, and virtually certain that this contributed to the grounding.

Continuing the transit at 8 kt after the plotter froze

The plotter froze at 18:07:06. The AIS track records no reduction of speed after that time. The vessel held 8.4 kt to 18:08, and the engine recorder shows no throttle movement before 18:11:52. The fall in speed over the ground to 7.6 kt by 18:11 is accounted for by the drag of shoaling water beneath the hulls, not by any action on the bridge.

The bridge audio records the master's attention moving to the search for the No. 4 beacon at 18:08:31, 85 seconds after the freeze. It records no consideration of stopping at any point.

The conditions favoured stopping. There was no other traffic in the channel and the water was flat. The vessel's trials figure for a crash stop from 8 kt is 92 m. The vessel crossed the channel's western edge 114 seconds after the freeze. A stop at any point in that window would have left it in the channel rather than on the bank.

The master gave his account in interview and in his safety report. He did not consider stopping. He went straight to the visual transit because, in his mind, the visual transit is the backup. He also said he would have stated beforehand that he would stop in that situation.

That account matters to how this factor is read. The master had about 4,100 transits of this channel over 11 years. He had made 26 of the 31 darkness transits since the winter timetable began. His last route check, on 14/02/2026, was satisfactory. An experienced master who would, in the abstract, have said he would stop did not think of stopping when the moment came. That points beyond the individual to what the procedures did and did not prompt him to do. Both masters

interviewed said, independently, that the right answer was to stop and that nothing tells you to. The absence of a hold point is analysed under Procedures.

No lookout was on the bridge. At 18:04:14 the master sent the deckhand to collect fares, which is the normal practice on the route. Nothing in OPS-3 addresses lookout practice either way. From that point the master was searching for the beacon, steering and monitoring the vessel alone.

This decision sits downstream of a passage plan with no hold point in it and a beacon that could not be seen from where the master was looking. It is a contributing factor. It is not the explanation of the occurrence on its own.

It is virtually certain that the master continued the transit at about 8 kt after the plotter froze rather than stopping in the channel, and extremely likely that this contributed to the grounding.

The plotter fault at the change of master

The plotter's event log records watchdog resets with cause code 0x51 at 10:42 and 11:17 local on the morning of 07/07/2026. The unit recovered by itself on both occasions. At the handover at about 13:05 the outgoing master told the relieving master about both freezes, and the relieving master replied that it had done the same to him the month before.

Neither master entered the fault in the wheelhouse defect book. Neither reported it to the Operations Manager. No entry of any kind was made for 07/07/2026 before the occurrence.

Both accounts of the handover agree in every particular. Both masters volunteered the omission before they were asked about it.

This was an omission both masters made. Nothing in the operator's procedures required either of them to avoid it. The finding concerns the practice, which is analysed under Procedures, rather than the two individuals. The explanation lies in the procedure. It does not excuse the omission, but it does account for it.

The influence of the omission does not rest on the single entry that could have been made at 13:05. An entry made then would have been reviewed at the Operations Manager's weekly review, not that afternoon. On its own it would not have stopped TF18 that evening.

The influence rests on the pattern the omission belongs to. The maintenance system holds no plotter entry of any kind for 2026. The operator had no subscription to the manufacturer's bulletins. The firmware issued in 11/2024 that addresses cause code 0x51 had never been applied to this unit. Had the freezes been recorded, they would have been visible to the people who acquire firmware and who decide whether a vessel sails. The fault the master met at 18:07 was one the manufacturer had already addressed.

It is virtually certain that the plotter fault was passed on verbally at the change of master and not entered in the defect book, and likely that this contributed to the grounding.

Duty pattern and fatigue

The route is worked on a cycle of 12 days on and 2 days off. 07/07/2026 was the master's ninth consecutive day. His duty that day was a split, 05:40 to 08:40 and 13:10 to 19:20 local. That gave 9 hours 10 minutes of duty across a spread of 13 hours 40 minutes, with an unpaid break of 4 hours 30 minutes at home in the middle. Both figures were inside the operator's limits of 10 hours of duty and 14 hours of spread, and inside the state's crewing determination.

The master recorded 6 hours 15 minutes of sleep the night before. He recorded it in the fatigue diary the operator asks masters to keep but does not collect. In interview he described himself as tired, said this was normal for a split duty, and said the roster had been the same for eleven years and he knew where he stood with it.

The case for influence, taken at its highest, is that this combination sits in the range where degraded decision making is expected. On that case, the decision that was not made, to stop when the plotter froze, is the kind of decision fatigue degrades.

The recorded evidence does not support that mechanism on the night:

- The master held the centreline to within 3 m for the first two minutes in the channel.
- His response to the shallow alarm was 4 seconds, against a fleet figure of 6 to 9 seconds.
- His radio work after the grounding was complete and correct at the first attempt.
- His decision not to refloat was made in about thirty seconds and was correct.

The one decision fatigue would have to work through is the failure to consider stopping. Both masters said independently that nothing in any document asks a master to consider it. The outgoing master, who was not fatigued, described the same understanding of the handover and the same difficulty with the beacon. The investigation therefore did not find that fatigue degraded a decision the master was never prompted to make.

The exposure remains. The winter timetable places the last darkness transit of the day at the end of a split duty whose spread comes within twenty minutes of the operator's own limit. That transit falls on the ninth to twelfth consecutive day of the cycle. The operator asks for a fatigue diary that it does not read. No relief master is rostered on the route. This exposure recurs on every darkness transit of the timetable.

It is virtually certain that the master was on the ninth consecutive day of a split duty with a spread of 13 hours 40 minutes. It is unlikely that this influenced the occurrence, and likely that this roster pattern increased risk.

The response after contact

The master stopped both engines at 18:12:31, 27 seconds after contact, and did not restart them. He addressed the passengers 36 seconds after the impact. His pan pan at 18:13:04 stated "I am not attempting to refloat" as a decision already made. He then had the No. 2 starboard void sounded, reported approximately 75 mm of ingress to Port Control at 18:17:05, and had the void sounded every ten minutes after that.

In interview he gave his reasons as the falling tide, the unknown bottom, the thirty-eight people aboard, and the fact that the vessel could not sink while it sat on the bank.

The survey bears out that judgement. The vessel was holed, with a 180 mm split in the starboard bottom plating in way of the No. 2 void. Ingress had reached approximately 260 mm by the time the vessel refloat at 00:20. Backing off the bank would have taken that compartment into 3.2 m of water, with the ingress unassessed, at night, on a falling tide, with the passengers still aboard.

The same analysis that finds the master should have stopped when the plotter froze also finds that his actions after the impact were correct and promptly taken.

It is virtually certain that the master stopped the engines and did not attempt to refloat, and very likely that this reduced the risk to the vessel and those aboard.

Environment

Darkness and the visual conditions

The transit was made 35 minutes after the end of civil twilight. The moon was a waning crescent, 2 per cent illuminated, which had risen at 06:12. Visibility exceeded 10 km. Cloud was few to scattered at 3,500 ft, and the sea in the channel was slight, with wind ripple only.

This was not reduced visibility in any ordinary sense. However, every visual reference available to the master was an artificial light. When the plotter failed, his ability to fix the vessel's position depended entirely on whether a particular light could be seen from where he was. As set out under Procedures, the reduced-visibility provision at OPS-3 3.6 did not apply on these terms.

The No. 4 beacon was not visible from the channel

The investigator requested a range and visibility check, which was made on 18/07/2026 from the port workboat, on the inbound track and at the same time of night. It found that the No. 4 beacon cannot be seen at all from the channel between 0.84 nm and 0.34 nm north of it.

Over that stretch the sightline passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. The canopy measured 5.8 m, against a lantern height of 4.2 m above chart datum. The light is visible beyond 0.84 nm, where the sightline passes clear of the islet's northern edge, and again inside 0.34 nm, where it opens past the islet's southern edge.

The bridge audio is consistent with that measurement. The vessel was 0.68 nm north of the beacon when the plotter froze. It was 0.47 nm north when the master asked "Where's four" at 18:08:31, and 0.25 nm north when he called "There's four" at 18:10:02, 91 seconds later. The vessel crossed the channel's western edge between those two remarks.

The master began searching inside the screened arc. He was still inside it when the vessel left the channel. By the time the light opened past the islet and he called it, the vessel was 39 m west of the centreline and outside the dredged channel. The master was not failing to look. Over the stretch that mattered, there was nothing to see.

Both masters interviewed raised the beacon without being asked. Neither had ever reported it.

Two limits on this finding are carried through. First, the measurement was made eleven days after the occurrence, and the canopy was not surveyed on the night itself. Second, the vessel had already begun to set west before the master started looking for the beacon. Its absence therefore explains why the set went uncorrected, not why it began.

With the plotter frozen, the beacon was the only means of fixing available to the master. It could not be seen for the whole period in which the vessel was set west.

It is extremely likely that the No. 4 beacon was not visible from the channel over the half mile before the vessel left it, and very likely that this contributed to the grounding.

Foreshore floodlighting behind the beacon (unresolved)

The investigation examined whether new floodlighting on the Kanooka Heads foreshore reduced the master's ability to pick out the No. 4 beacon. It could not settle the question either way.

Evidence supporting an effect. In interview the master described looking into a wall of light, and said the beacon was negligible against the town and the new car park. The outgoing master, interviewed separately, described the beacon as washed out altogether from further up the channel.

The foreshore redevelopment was commissioned on 22/05/2026, ten days before the winter timetable began. It includes 14 LED floodlights on 12 m columns at the new boat ramp and car park. The survey from the port workboat on 18/07/2026 found that each floodlight presents about 4,800 candela toward the channel, and none is shielded to seaward. From 0.5 nm north of the No. 4 beacon, the array lies behind the beacon within 4 degrees of the same bearing and within 1.5 degrees of the same elevation. The array is continuously lit, while the beacon flashes green for 0.4 seconds every 2.5 seconds.

The redevelopment was approved by the Council's planning unit. The Port Authority's aids to navigation officer was not consulted, and no assessment of the lighting against the channel's aids was required or made.

Evidence against an effect. Over the stretch where the master was actually looking, the beacon was screened by the mangrove canopy. It would not have been visible against a black background either. The one point at which the alignment was measured, 0.5 nm north of the beacon, lies inside the screened arc. Once the light did open past the islet, the master picked it up with the floodlights in the same place and no dimmer.

Why the question could not be settled. No measurement exists of what could be seen on the night, only the survey eleven days later. Nothing establishes a detection threshold for a 0.4 second green flash against that background. The alignment is real, and the masters' accounts are consistent and independent. However, while the canopy screens the beacon over the same stretch, the effect of the floodlights cannot be separated from the effect of the mangroves.

What would resolve it. Settling the question would need a detection measurement for the beacon against the floodlit background along a part of the inbound track where the beacon is not screened. That may not be possible until the sightline through the Weerona Islet canopy is cleared.

The investigation considered whether the existence of this effect was established and assessed it as about as likely as not. The matter therefore remains unresolved. It is not carried into the findings. Independently of how it is resolved, the absence of any assessment of the foreshore lighting against the channel's aids indicates that such an assessment is warranted.

Equipment

The chart plotter froze and did not recover

Three independent sources place the freeze at 18:07:06:

- The plotter's internal event log records the display process not responding at that time, with no accompanying reset.
- The bridge audio records "Oh, come on" at 18:07:11, sounds consistent with repeated screen taps at 18:07:26, and "frozen again" at 18:07:44.
- The master's account is that the chart picture stayed on the screen with the vessel symbol stationary, and the display would not accept touch input.

The cause code was 0x51, the same code logged at 10:42 and 11:17 that morning. On the earlier occasions the unit had reset itself. On this occasion it did not.

An approved electronics contractor bench tested the unit for 46 hours on 21/07/2026 with the same antenna and cabling. The fault did not recur and no defect was found. The contractor's report notes that cause code 0x51 is known to the manufacturer and is addressed by firmware issued in 11/2024. That firmware had not been applied to this unit, which was running firmware issued 03/2024.

The bench test result bears on what caused the freeze, not on whether the display froze during the transit. The test could not reproduce the fault, so the cause was not confirmed by testing. The match between the logged cause code and the manufacturer's known issue is the evidence that links the freeze to the firmware the operator had not received.

The freeze was the point at which the transit departed from the passage plan. From then on the master navigated without the primary means of position fixing that the plan told him to use. The vessel had held the centreline while the plotter worked. It began leaving the centreline at the moment of the freeze, and it crossed the channel edge 114 seconds later.

It is virtually certain that the chart plotter display froze during the channel transit and did not recover, and extremely likely that this contributed to the grounding.

The echo sounder shallow alarm

The echo sounder's shallow alarm was set to 0.9 m under the keel. That value was written into the setup when the unit was commissioned in 2019 and has not been changed.

The alarm sounded once on the voyage, at 18:11:48. The master moved both throttles to astern together 4 seconds later, and contact followed at 18:12:04, 16 seconds after the alarm. At 7.1 kt the vessel covered 58 m in that time. The vessel's own trials figure for a crash stop from 8 kt is 92 m. Astern thrust was developing at the moment of contact but had not reached full.

The surveyed bank gradient across the contact section is about 1 in 70. At that gradient, the alarm could not have sounded appreciably earlier at the 0.9 m setting.

The equipment did exactly what it was set to do. The problem was the combination of the setting and the speed. A 0.9 m alarm on a 1 in 70 gradient at about 8 kt gives less warning than the distance the vessel needs to stop. A setting that gave more warning, or a transit speed at which 16 seconds was enough, would have changed the outcome.

The alarm was the vessel's last defence, and at that setting and speed it could not have prevented contact once the vessel was on the bank.

It is virtually certain that the shoaling was not detected until 16 seconds before contact, and extremely likely that this contributed to the grounding.

The forward saloon lifejacket locker

This matter came to light through the deckhand's account. While issuing lifejackets after the grounding, she had to ask three passengers standing at the kiosk counter to move. She then had to pass behind the counter through a narrow gap to reach the forward saloon locker.

The stowage survey on 16/07/2026 found the following:

- The locker holds 40 adult lifejackets and is mounted in the bulkhead behind the kiosk servery counter.
- The locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter through a 0.6 m gap at its inboard end.
- With three people standing at the counter, as there were on the night, that gap is the only route.
- The muster placard in the forward saloon directs passengers to "the locker forward on the port side". That is the locker in question, and it cannot be seen from the seats.

The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear. The counter was fitted during a 2019 interior refit carried out under a minor modification notice. That notice did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.

The locker did not affect the grounding or its outcome. The vessel would have grounded wherever the locker was. All 38 passengers were in lifejackets by about 18:22, within about eight minutes of the master's instruction, and all were ashore within 74 minutes of the impact. The vessel was aground, upright, stable and not sinking, and the muster was unhurried. It took somewhat longer in the forward saloon than it otherwise would have.

The importance of the arrangement lies in other circumstances, such as:

- a fire

- a risk of capsize
- a night evacuation into the water
- a deckhand injured in the impact.

In any of these, a locker that passengers cannot reach without crew help separates a saloon whose occupants can equip themselves from one that has to wait. The underlying gap is the modification route rather than the counter itself. An interior refit changed a safety equipment arrangement, and nothing in that route asked whether it had done so. The same arrangement is repeated across the operator's class of 24 m vessels.

It is virtually certain that the forward saloon lifejacket locker could not be reached from most of the saloon it serves. It is unlikely that this influenced the occurrence, and very likely that it increased risk.

Procedures

No night passage plan and no hold point if the primary means of fixing is lost

OPS-3 3.4 requires the Weerona Channel transit to be made "on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon".

Appendix A, issue 2, dated 02/2023, is the current Weerona Channel passage plan. It names the chart plotter as the primary means of position fixing. As the secondary means it names the "visual transit of the No. 4 beacon and the town wharf leading lights". It lists the No. 4 beacon at a range of 6 M. It closes with the note that the plan is written for the daylight transit and that "Night transits are not scheduled on this route." That line is the plan's only reference to darkness.

The plan therefore had four gaps:

- There is no night passage plan for the Weerona Channel.
- There is no guidance anywhere on what to do if the plotter is lost during the transit.
- No abort criterion or hold point is named anywhere in the plan.
- The secondary means of fixing was, in practice, only one light at the point where the plotter failed.

The last point needs explaining. The town wharf leading lights bear 195 degrees, but they lie 2.1 nm beyond the No. 6 beacon. They do not resolve as two lights until the vessel is inside the channel's southern third. At the point in the transit where the plotter was lost, the No. 4 beacon was the whole of the secondary means. As analysed under Environment, that beacon could not be seen over the relevant stretch.

OPS-3 3.6 is the nearest provision to a stopping rule. It requires the master, "in reduced visibility", to reduce speed so the vessel can be stopped within the distance in which the channel can be seen, and to consider anchoring or returning to open water. Visibility on the night exceeded 10 km, so the condition that triggers that provision was not met. Nothing in the manual addressed the situation the master was in: good visibility and no means of fixing.

The plan was the operator's own defence against exactly this sequence, and that defence had never been written. A hold point would have cost nothing on a flat night with no traffic. It would have been available for the whole 114 seconds between the freeze and the vessel crossing the channel edge. Every master asked said they would have used one.

There is a limit on this finding. A master who did not think of stopping unprompted might not have applied a written rule under the same pressure.

It is virtually certain that there was no night passage plan for the channel and no hold point if the primary means of fixing was lost, and very likely that this contributed to the grounding.

The change-of-master defect brief

OPS-9 sets out three relevant provisions.

- Section 9.1 requires immediate reporting only of a defect "which renders the vessel unseaworthy or which affects a statutory item of equipment". The chart plotter is not a statutory item on a class 1D vessel on this route, even though it is the primary means of position fixing named in the passage plan.
- Section 9.2 requires all other defects to be entered in the defect book "at the end of the shift in which they are found", for weekly review by the Operations Manager.
- Section 9.3 requires the outgoing master to brief the relieving master on "the state of the vessel, any defect, and anything else affecting the conduct of the service".

Nothing in OPS-9 requires the brief to be written down. Nothing requires the relieving master to confirm that a defect mentioned in the brief has been entered in the book. For an intermittent fault on a non-statutory item, the only trigger for an entry is the end of the shift in which it is found. For a fault that has come good by then, that is no trigger at all.

Both masters described the same understanding independently: the defect book is for what is broken at handover, and what has come good is spoken about. The outgoing master believes the plotter has frozen three or four times on this vessel in the past year, to him and to others, and that none of those freezes was ever written up. The maintenance system holds no plotter entry of any kind for 2026.

There are two readings of this rule's influence.

The reading that it contributed. The rule produced exactly the outcome it permits. An intermittent fault on the vessel's primary means of position fixing was known to two masters on the day but invisible to everyone ashore. The relevant consequence is not the single entry missed at 13:05. It is that repeated freezes over a year never reached anyone who could act. As a result, the firmware issued in 11/2024 was never applied, and the master met the fault on the night with no reason to treat it as more than a nuisance.

The reading that it did not. An entry made at 13:05 would have been reviewed at the end of that week, not that afternoon, and the vessel would have sailed TF18 either way. The path from a written entry to a different outcome on the night runs through a same-day decision that nobody in the system was required to make. On this reading, the handover rule is a real finding about the fleet but not a cause of this grounding.

The investigation weighed both readings and accepted the first on the balance of the evidence, at a lower level of confidence than most of the other influence findings. The weight rests on the year of unrecorded freezes and the firmware that was never obtained, not on the events of the afternoon.

It is virtually certain that the change-of-master defect brief was verbal, with nothing to check that a defect mentioned in it was recorded, and likely that this contributed to the grounding.

Organisation

The winter timetable and the route risk assessment

RRA-07 issue 2, dated 03/2023, is the current route risk assessment for the Weerona Channel. No issue 3 exists. Among the hazards it lists are Boolarong Bank to the west of the channel, a channel

width of 40 m against a vessel beam of 8.2 m, and a "single unlit reach between the No. 2 and No. 4 beacons".

It also imposes an operating restriction: "daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan."

CHG-2026-014 introduced the winter timetable from 01/06/2026, moving TF18 from 16:15 to 17:35 to meet the afternoon coach connection at the town wharf. It recorded the change as follows:

- Assessment: "timetable change only".
- Hazards: "No new hazard identified".
- "Route risk assessment review required": "No".
- Approval: the Operations Manager, on 22/04/2026.

The daylight-only restriction is not referenced anywhere in CHG-2026-014. RRA-07 was not reopened. No night passage plan was written. The General Manager Operations' approval, which the restriction requires for a night transit, was neither sought nor given. The change was approved one level below the person the restriction names. From 01/06/2026, TF18 crossed the channel after the end of civil twilight on every day of the timetable, 31 times before the occurrence.

The change process asked whether the change was new. It did not ask whether the existing assessment still held. As a result, a live restriction was overridden without anyone deciding to override it.

A review of RRA-07 against a night operation would have reached the unlit reach between the No. 2 and No. 4 beacons, which the assessment already named as a hazard. It would also have required a night passage plan before the first service ran.

Two matters temper that conclusion. First, a review might have produced a plan that said no more than the daylight one. Second, the Operations Manager's understanding of what the restriction required of him has been given only verbally and has not been taken as a formal interview. Why the change was approved in this way is therefore less firmly established than the fact that it was.

It is virtually certain that the winter timetable moved the last service into darkness without reopening the route risk assessment, and likely that this contributed to the grounding.

The solar conversion of the No. 4 beacon

In 11/2025 the No. 4 beacon was converted from mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal range of 3 nautical miles, against 6 for the lantern it replaced. The character, Fl G 2.5s, did not change.

A notification was lodged with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued, and no operator using the channel was told. The investigator's measurement on 18/07/2026 gave a luminous range of 2.6 nm.

On the night, the master was working to a passage plan that listed the No. 4 beacon at 6 M. The beacon in front of him had been a 3 M unit for eight months, and its measured range was 2.6 nm.

A weaker light is harder to acquire, and the master spent 91 seconds failing to acquire this one. That mechanism is limited by the screening, however. The stretch where he was looking is also the stretch where the light was screened entirely, and a screened light is not made visible by being brighter. The influence of the reduced range is therefore real but bounded.

This finding does not depend on the foreshore floodlighting. As discussed under Environment, whether the floodlighting affected detection was not resolved.

It is virtually certain that the beacon was converted to a solar lantern with its nominal range halved and that no operator was told, and likely that this contributed to the grounding.

Deferral of the clearing of the beacon's sightline

The Port Authority's aids to navigation register shows the following history:

- **09/2023.** Work order WO-118 was raised for mangrove regrowth on Weerona Islet screening the No. 4 beacon from the northern approach, between about half a mile and a quarter mile. It recommended trimming to 2 m on a three-yearly cycle, at an estimated \$18,500.
- **2023, 2024 and 2025 works rounds.** The work was deferred at each round as unfunded and referred to the next round.
- **03/2026.** Work order WO-176 recorded the same sightline as now fully closed on the northern approach.

The Authority's file note against WO-176 explains why the work kept losing out. The islet is 240 m from the beacon, outside the 50 m vegetation scope in schedule 2.4 of the services agreement. The clearing therefore competes in the general waterways vote with boat ramps and pontoons, rather than being funded as aids to navigation work.

The 2019 aerial survey gave the canopy height as 2.1 m. On 18/07/2026 it measured 5.8 m, against a lantern height of 4.2 m. The canopy grew past the lantern's height at some point between 2019 and 2023, and the Authority's own register has described the screening since 09/2023.

The monthly checks did not detect the problem. The beacon is logged as observed and operating in every month from 01/2024 to 07/2026, most recently on 02/07/2026. Those checks were made from the town wharf, 2.1 nm south of the beacon, from where Weerona Islet is not in the line of sight. The beacon was correctly observed to be lit from the one direction in which the screening could not be seen. No defect has ever been recorded against it. No notice about it has ever been issued to a waterway user. The register has never been provided to Tarrakin Ferries.

The screened arc measured on 18/07/2026 runs from 0.84 nm to 0.34 nm north of the beacon, and the vessel left the channel inside it. This deferral is the physical reason the light was not where the master expected it: it was not dim or defective, but absent. The finding is qualified only by the possibility that the master might not have picked up the light in that minute and a half even if it had been there to see.

It is virtually certain that the clearing of the beacon's sightline was deferred unfunded for three works rounds, and very likely that this contributed to the grounding.

The aids to navigation code and the channel's status

Section 4.2 of the Waterways Aids to Navigation Code WAN-2 applies three duties to aids on a "declared shipping channel":

- an obligation to record availability
- an availability target of "not less than 99.0 per cent measured annually"
- an obligation to "notify a change to the aid's character, position or range by notice to mariners before the change takes effect".

The Weerona Channel is not a declared shipping channel. Section 4.3 applies instead. It requires only that the responsible authority "be able to demonstrate that the aid is inspected at intervals it has determined and recorded".

The services agreement between Kanooka Shire Council and the Port Authority sets those intervals. It requires a quarterly daylight inspection and a monthly observation from the town wharf. It limits

vegetation work to within 50 m of an aid. It states that there is "no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid."

The Authority was not in breach of either instrument. The result of the code's threshold is that a 40 m channel used by a 96-passenger ferry has beacons that nobody is required to measure, record or report on.

There are two readings of this factor's influence.

The reading that it contributed. Every mechanism that would have carried the beacon's condition to the people using it exists in the code and stops at this channel's boundary: a measured range, a recorded availability figure and a notice to mariners. In their place, the only channel of information was two masters telling each other informally. The notification duty at 4.2 is the one mechanism in the evidence that would have put the range change in front of the operator. The operator's passage plan still carried the old 6 M figure on the night.

The reading that it did not. Nothing in the code stopped the Authority from notifying operators, trimming the mangroves or measuring the light. It did not do so for reasons of money and priority, not because the code allowed it. A 99 per cent availability target would not have caught this problem at all, because the lantern was lit and working throughout. The difficulty was that the light could not be seen, which no availability figure measures. On this reading the code's scope is background, not cause.

The investigation accepted the first reading on the balance of the evidence, at the lowest confidence of any influence finding in this analysis. The second reading is not dismissed and is presented here for that reason.

It is virtually certain that aids to navigation on this waterway carry no availability standard and no duty to notify users, and likely that this contributed to the grounding.

Hypotheses tested and excluded

Two hypotheses were tested and ruled out by measurement.

Lower water level than predicted. The water level over Boolarong Bank was not below prediction, which makes this hypothesis very unlikely. The Port Authority's gauge read 0.91 m at 18:12 against a predicted 0.94 m, and its largest departure from prediction in the twelve hours to 20:00 was 0.07 m. With the bed at 0.50 m below chart datum, there was 1.41 m of water at the contact point. That compares with a static draught of 1.35 m and a calculated squat of 0.18 m at 6.8 kt. The vessel touched because it was 54 m outside a channel dredged to 3.2 m.

Steering or propulsion failure. A steering or propulsion failure was also found very unlikely, on two independent grounds. The first is the recorded data:

- Both engines ran within 12 rpm of each other from 18:03 to 18:11:52, with no alarm, no excursion and no throttle movement.
- Steering hydraulic pressure held between 96 and 99 bar, within its normal band of 90 to 105 bar.
- Rudder angle followed helm demand within 1.5 degrees.
- The rate of turn describes a heading applied and held, not a sheer.

The second is the post-occurrence examination. The whole steering system was removed and function tested on 12/07/2026, and every component was serviceable, with no sign of a pre-existing defect.

Findings summary

A finding stated without a likelihood term was established as fact.

Contributing factors

- The winter timetable change moved the last service, TF18, into darkness without the route risk assessment for the Weerona Channel being reopened, and this likely contributed to the grounding. (Safety Issue)
- It was virtually certain that the operator had no night passage plan for the channel and no hold point for when the primary means of position fixing was lost, and this very likely contributed to the grounding. (Safety Issue)
- Under the operator's procedures the change-of-master defect brief was verbal, with nothing to check that what was said was written down, and this likely contributed to the grounding. (Safety Issue)
- At the change of master at about 13:05, the chart plotter fault was passed on verbally and not entered in the defect book, an omission that likely contributed to the grounding.
- During the channel transit the chart plotter display froze at 18:07:06 and did not recover, and this very likely contributed to the grounding.
- After the plotter froze, the transit continued at eight knots rather than the vessel stopping in the channel, which very likely contributed to the grounding.
- Because the Weerona Channel was not a declared shipping channel, aids to navigation on this waterway carried no availability standard and no duty to tell users anything, and on the balance of the evidence this likely contributed to the grounding. (Safety Issue)
- When the No. 4 beacon was converted to solar in 11/2025 its nominal range was halved and no operator was told, a condition that likely contributed to the grounding. (Safety Issue)
- Clearing of the No. 4 beacon's sightline through the mangrove canopy on Weerona Islet was deferred unfunded for three works rounds, and this very likely contributed to the grounding. (Safety Issue)
- It was very likely that the No. 4 beacon was not visible from the channel over the half mile before the vessel left it, and very likely that this contributed to the grounding.
- The vessel left the dredged channel and tracked onto Boolarong Bank, and this contributed to the grounding with virtual certainty.
- Shoaling was not detected until sixteen seconds before contact, with the echo sounder's shallow alarm set at 0.9 m under the keel against a crash stop distance of 92 m from 8 kt, and this very likely contributed to the grounding.

Other factors that increased risk

- The forward saloon lifejacket locker was not reachable from most of the saloon it serves, because a kiosk counter fitted in a 2019 refit under a minor modification notice blocked the locker face from 22 of the 34 forward saloon seats, and although it did not influence this occurrence this very likely increased risk. (Safety Issue)
- On the day of the occurrence the master was on day nine of twelve, on a split duty across a thirteen hour forty minute spread that was within every applicable limit, and although this was not found to have influenced the occurrence it likely increased risk on the route's winter darkness transits.

Other findings

- It was very unlikely that a water level below prediction reduced the depth over Boolarong Bank, and this hypothesis was examined and ruled out on the Port Authority's gauge reading of 0.91 m at 18:12 against a predicted height of 0.94 m.
- It was very unlikely that a steering or propulsion failure caused the vessel to sheer off track, and this hypothesis was examined and ruled out because the recorded engine, steering gear and rudder data were normal throughout and every steering component was found serviceable when tested on 12/07/2026.
- After contact the master stopped both engines and made no attempt to refloat on a falling tide, a decision that very likely reduced risk because the vessel was holed in the compartment it was resting on and 38 passengers were aboard.

Appendices

Appendix A: Test summary

Each factor examined in this investigation was tested for whether it existed and whether it influenced the occurrence. Where a factor existed but did not influence the occurrence, it was also tested for whether it was important to safety. The table shows the result of each test, the likelihood behind it, and the category that followed. Where a test is shown as not tested, it was not applied.

Factor	Lane	Existence	Influence	Importance	Category
Aids to navigation on this waterway carry no availability standard and no duty to tell users anything	External Influences	Pass (virtually certain)	Pass (likely)	Not tested	Contributing factor (Safety Issue)
Beacon converted to solar, nominal range halved, and no operator was told	Organisational Influences	Pass (virtually certain)	Pass (likely)	Not tested	Contributing factor (Safety Issue)
Clearing of the beacon's sightline deferred unfunded for three works rounds	Organisational Influences	Pass (virtually certain)	Pass (very likely)	Not tested	Contributing factor (Safety Issue)
Winter timetable moved the last service into darkness without reopening the route risk assessment	Organisational Influences	Pass (virtually certain)	Pass (likely)	Not tested	Contributing factor (Safety Issue)
No night passage plan for the channel, and no hold point if the primary means of fixing is lost	Risk Controls	Pass (virtually certain)	Pass (very likely)	Not tested	Contributing factor (Safety Issue)
Change-of-master defect brief is verbal, with nothing checking that what is said gets written down	Risk Controls	Pass (virtually certain)	Pass (likely)	Not tested	Contributing factor (Safety Issue)
No. 4 beacon not visible from the channel over the half mile before the vessel left it	Local Conditions	Pass (extremely likely)	Pass (very likely)	Not tested	Contributing factor
Chart plotter display froze during the channel transit and did not recover	Local Conditions	Pass (virtually certain)	Pass (extremely likely)	Not tested	Contributing factor
Plotter fault passed on verbally at the change of master and not entered in the defect book	Individual Actions	Pass (virtually certain)	Pass (likely)	Not tested	Contributing factor
Transit continued at eight knots after the plotter froze, rather than stopping in the channel	Individual Actions	Pass (virtually certain)	Pass (extremely likely)	Not tested	Contributing factor

Vessel left the dredged channel and tracked onto Boolarong Bank	Occurrence Events	Pass (virtually certain)	Pass (virtually certain)	Not tested	Contributing factor
Shoaling not detected until sixteen seconds before contact	Occurrence Events	Pass (virtually certain)	Pass (extremely likely)	Not tested	Contributing factor
Forward saloon lifejacket locker not reachable from most of the saloon it serves	Risk Controls	Pass (virtually certain)	Fail (unlikely)	Pass (very likely)	Other factor that increased risk (Safety Issue)
Day nine of twelve, on a split duty across a thirteen hour forty minute spread	Local Conditions	Pass (virtually certain)	Fail (unlikely)	Pass (likely)	Other factor that increased risk
Engines stopped and no attempt made to refloat on a falling tide	Individual Actions	Pass (virtually certain)	Pass (very likely)	Not tested	Positive event that reduced risk
New foreshore floodlighting behind the beacon reduced the master's ability to pick it up	Local Conditions	Inconclusive (about as likely as not)	Not tested	Not tested	Unresolved, no finding made
Water level below prediction reduced the depth over Boolarong Bank	Local Conditions	Fail (very unlikely)	Not tested	Not tested	Excluded hypothesis
Steering or propulsion failure caused the vessel to sheer off track	Occurrence Events	Fail (very unlikely)	Not tested	Not tested	Excluded hypothesis

For the floodlighting matter, the existence test was examined and could not be settled. The investigation found a real alignment between 14 unshielded floodlights at the new boat ramp and the No. 4 beacon from 0.5 nm north of it, and both masters independently described the beacon as hard to see against that background. However, the only point at which the alignment was measured lies inside the arc where the mangrove canopy screened the beacon entirely, and the master acquired the light at 0.34 nm with the floodlights unchanged. No measurement exists of what could be seen on the night, and no detection threshold has been established for a 0.4 second green flash against that background. The matter may not be answerable while the canopy remains in place.

Appendix B: Excluded hypotheses

Water level below prediction reduced the depth over Boolarong Bank

This hypothesis was tested because the vessel grounded three hours into a falling tide after a westerly wind had been blowing for two days, and a set-down of a few tenths of a metre would have mattered. It was found to be very unlikely, on measurement rather than argument. The Port Authority's gauge at Kanooka Heads read 0.91 m at 18:12, against a predicted height of 0.94 m for that minute, so the water was 0.03 m below the tide table. The gauge's largest departure from the table in the twelve hours to 20:00 was 0.07 m. The gauge and the tide table are independent of each other and of the vessel. With the bed at the contact point surveyed at 0.50 m below chart datum, there was 1.41 m of water over it. The vessel's static draught was 1.35 m and its calculated

squat was 0.18 m at 6.8 kt. The vessel touched because it was 54 m outside a channel dredged to 3.2 m, not because the water was lower than predicted.

Steering or propulsion failure caused the vessel to sheer off track

This hypothesis was tested because a five degree heading change followed by a steady set out of the channel is what a jammed rudder or a lost engine would look like from outside the vessel. It was found to be very unlikely, on two independent grounds. The engine recorder showed both engines at 1,780 and 1,784 rpm, within 12 rpm of each other, from 18:03 to 18:11:52, with no alarm, no excursion and no throttle movement. Steering gear hydraulic pressure held continuously at 96 to 99 bar against a normal band of 90 to 105 bar, and rudder angle followed helm demand within 1.5 degrees throughout. The rate of turn peaked at 2.1 degrees per minute and returned to zero, which is consistent with a heading applied and then held rather than a sheer. Separately, the whole steering system was removed and function tested on 12/07/2026. Every component was serviceable, with no internal leakage and no sign of a pre-existing defect. The damage to the starboard shaft, propeller and rudder dated from the contact. The bridge audio contained no remark about steering or the engines at any point.

Appendix C: Sources of evidence

Interviews and reports from those involved

- The master, interviewed on 14/07/2026. The interview was recorded and transcribed.
- The deckhand, interviewed on 14/07/2026. The interview was not recorded, at her request.
- The outgoing master, interviewed by telephone on 23/07/2026. The interview was not recorded. Notes were read back to him and agreed.
- The master's safety report SR-2026-0288, timed 18:52 on 07/07/2026.
- The deckhand's safety report SR-2026-0291.

Recorded data and audio

- Port Control AIS receiver log and vessel recorder readout AIS-2026-0093-R1, issued 15/07/2026, covering the AIS track, engine monitoring recorder and steering data for service TF18.
- AUD-2026-0093, VHF channels 12 and 16 and bridge camera audio, 17:32 to 19:26 local, transcribed 16/07/2026 and verified by the Port Authority's traffic officer.
- Chart plotter internal event log, recovered 09/07/2026.
- Echo sounder setup and alarm log, recovered 09/07/2026.
- MET-TIDE-0093-070726, retrieved 09/07/2026, covering tide predictions and the gauge reading, weather observations, and light and moon conditions.

Examinations, surveys and tests

- Survey of the grounding position and bed levels, conducted on 08/07/2026.
- Survey and slipping report for MV Tarrakin Star, 09/07/2026 to 14/07/2026, including the steering gear examination and function test on 12/07/2026.
- Forward saloon lifejacket stowage survey, MV Tarrakin Star, 16/07/2026.
- Range and visibility check of the No. 4 beacon from the port workboat on the inbound track, at night, 18/07/2026.
- Kanooka Heads foreshore lighting survey from the port workboat, 18/07/2026.

- Contractor bench test report on the chart plotter, 21/07/2026.

Documents

- Waterways Aids to Navigation Code WAN-2, section 4.
- Aids to navigation services agreement between Kanooka Shire Council and Kanooka Heads Port Authority, schedule 2.
- Kanooka Heads Port Authority aids to navigation register and register extract for the No. 4 beacon, copied 19/07/2026, including work orders WO-118 and WO-176.
- The Port Authority's 2019 aerial survey of Weerona Islet.
- Tarrakin Ferries Operations Manual OPS-3 revision 6 (02/2023), passage planning, and appendix A issue 2, the Weerona Channel plan.
- Tarrakin Ferries Operations Manual OPS-9 revision 4 (08/2024), defect reporting and rectification.
- Route Risk Assessment RRA-07 issue 2 (03/2023) and change record CHG-2026-014 (approved 22/04/2026).
- Crew records, certificates and roster for 07/07/2026, Tarrakin Ferries, copied 13/07/2026.
- Vessel maintenance history and wheelhouse defect book, MV Tarrakin Star, copied 10/07/2026.

Appendix D: Investigation activity

- 07/07/2026: The investigation was notified by the master's safety report SR-2026-0288, before the operator's notification was raised. The occurrence was assessed as a serious marine incident warranting investigation, and the investigation was opened and its scope set.
- 08/07/2026: The vessel had been refloated overnight and secured at the town wharf. The chart plotter and echo sounder heads were removed and sealed. The Port Control AIS log and VHF recordings were requested, and the bridge camera was downloaded before the vessel was slipped. The master and the deckhand were stood down and tested, with negative results. Evening Muntari services were suspended by the General Manager Operations.
- 10/07/2026: The survey of the grounding position and bed levels, conducted on 08/07/2026, was received. The steering gear was removed for function testing. The deckhand's safety report SR-2026-0291 was received, and the forward saloon lifejacket stowage was added to the scope of the investigation on the strength of it.
- 14/07/2026: The master was interviewed and the interview recorded. The deckhand was interviewed, without recording at her request. Both raised the No. 4 beacon. A night range and visibility check of the beacon from the port workboat on the inbound track was requested.
- 19/07/2026: The range and visibility check had been completed, finding that the beacon was not visible from the channel between 0.84 nm and 0.34 nm north of it, and that the solar conversion of 11/2025 had halved its nominal range with no notice to any operator. The foreshore lighting was surveyed the same night. The aids to navigation register was obtained, showing that WO-118 had been deferred for three works rounds.
- 23/07/2026: The outgoing master was interviewed by telephone. Both masters interviewed had raised the beacon unprompted, and neither had ever reported it. The investigation found that the operator's "navigation hazard" reporting category was

unused on this route and unknown to all of those interviewed. The beacon findings were raised with the Port Authority and the operator.

Distribution

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